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TOPIC: **“Causality Relationship between Interest Rate and
Stock Returns in India - An Analytical Study”**

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ABSTRACT

The present study tries to explore the causality relationship between interest rate and stock returns. Using monthly Daily data from January 2015 to December 2015. We have considered Weighted Lending rates as a proxy for the Interest Rate and BSE SENSEX for the study. The Series made stationary in order to apply the Granger Causality Test. The study reveals that there exists no causal relationship among the interest rate and stock returns through Granger causality technique. The study implies that the interest rate neither affects stock returns nor stock returns affect the interest rate. Thus, the present study empirically proves, stock market has no relation with the growth of interest rate in India and vice versa for selected period of time.

Keywords: Stock Returns, Lending Rates, Granger Causality Test ,Interest Rate.

Introduction

In the era of globalization and stabilization, one of the macroeconomic variables that have come into greater focus is the interest rate. This is consequent upon a strengthened integration of the domestic financial sector with the external sector. In India, the financial sector has undergone many changes since the stabilization program initiated in the early 1990s. With financial liberalization in the economy, flexibility has been imparted to the movement of interest rates. The relationships between stock market returns and interest rate has been examined by researchers as it plays important role in influencing a country's economic development (Aydemir and Demirhan, 2009). Interest rates are determined by monetary policy of a country according to its economic situation. High interest rate will prevent capital outflows, hinder economic growth and, consequently, hurt the economy as interest rates is one of the most important factors affecting directly the growth of an economy. The rational for the relationship between interest rate and stock market return are that stock prices and interest rates are said to be negatively correlated (French et. al.1987). Higher interest rate resulting from adverse monetary policy negatively affects stock market returns because higher interest rate reduces the value of equity and makes fixed income securities more attractive. On the contrary, lower interest rates resulting from expansionary monetary policy boosts stock market (Fama, 1981; Geske and Roll, 1983).

This is supposed to be a good sign for the economy as it is expected to enhance the efficiency in the financial system and thereby leading to the achievement of a higher growth rate. This policy stance is mostly backed up by the position of the monetarist and the financial liberalization school as opposed to the Keynesian school. According to the Monetarists, 'a flexible interest rate policy responds to the changes in the market conditions (demand and supply of credit), thereby enabling the economy to withstand and control the macroeconomic instabilities as an inflexible interest rate policy is prone to macroeconomic fluctuations' (King and Lin, 2005). A rise in the interest rates affects the valuation of the stocks. The rise in the interest rates raises the expectations of the markets participants, which demand better returns that commensurate with the increased returns on bonds. In a low interest rate regime, corporate are able to increase profitability by reducing their interest expenses. However in a rising interest rate regime, as interest expenses rise, profitability is affected. When interest rates rise, investors move from equities to bonds. Whereas when interest rates fall, returns on bonds fall while the returns on equities tends to look relatively more attractive and the migration of fund from bonds to equities takes place, and increasing the prices of equities. Though financial economists, policy makers and investors have long-attempted to understand dynamic interactions between interest rate and stock prices, the exact patterns of the

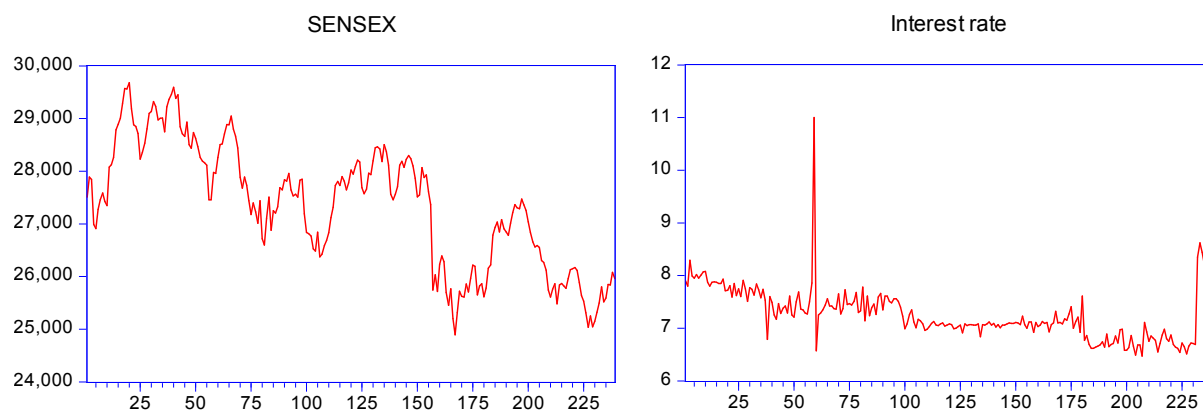
interactions remain unclear, the nature and strength of the dynamic interactions between them is of high interest and need to be evaluated empirically. Therefore, the researcher examines the dynamic relationship between interest rate and stock prices in order to identify the impact of interest rate changes on stock prices with special reference to Indian Stock Exchange.

Reasons for Interest rate Changes

- **Political short-term gain:** Lowering interest rates can give the economy a short-run boost. Under normal conditions, most economists think a cut in interest rates will only give a short term gain in economic activity that will soon be offset by inflation. The quick boost can influence elections. Most economists advocate independent central banks to limit the influence of politics on interest rates.
- **Deferred consumption:** When money is loaned the lender delays spending the money on consumption goods. Since according to time preference theory people prefer goods now to goods later, in a free market there will be a positive interest rate.
- **Inflationary expectations:** Most economies generally exhibit Inflation, meaning a given amount of money buys fewer goods in the future than it will now. The borrower needs to compensate the lender for this.
- **Alternative investments:** The lender has a choice between using his money in different investments. If he chooses one, he forgoes the returns from all the others. Different investments effectively compete for funds.
- **Risks of investment:** There is always a risk that the borrower will go bankrupt, abscond, die, or otherwise [default](#) on the loan. This means that a lender generally charges a risk premium to ensure that, across his investments, he is compensated for those that fail.
- **Liquidity preference:** People prefer to have their resources available in a form that can immediately be exchanged, rather than a form that takes time to realize.
- **Taxes:** Because some of the gains from interest may be subject to taxes, the lender may insist on a higher rate to make up for this loss.
- **Banks:** Banks can tend to change the interest rate to either slow down or speed up economy growth. This involves either raising interest rates to slow the economy down, or lowering interest rates to promote economic growth.
- **Economy:** Interest rates can fluctuate according to the status of the economy. It will generally be found that if the economy is strong then the interest rates will be high, if the economy is weak the interest rates will be low.

Theoretical Rational

Stock exchange and interest rate are two decisive factors of economic expansion of a country. The impacts of interest rate on stock exchange endow with imperative implications for monetary policy and government policy towards financial markets. Interest rates are determined by monetary policy of a country according to its economic state of affairs. High interest rate will put off capital outflows, hamper economic growth and, consequently, upset the economy as interest rates is one of the most vital factors affecting directly the development of an economy. The rational for the relationship between interest rate and stock market return are that stock prices and interest rates are said to be negatively correlated (French et. al.1987). Higher interest rate resulting from monetary policy negatively affects stock market returns because higher interest rate reduces the value of equity and makes fixed income securities more eye-catching. On the contrary, lower interest rates resulting from expansionary monetary policy boosts stock market (Fama, 1981; Geske and Roll, 1983). Financial theory states that movements in interest rates affect both the firm's expectations about future corporate cash flows and the discount rate employed to value these cash flows and, hence, the value of the firm. The impact of interest rate fluctuations on the market value of companies has received a great deal of attention in the literature, although much of the empirical research has focused on financial institutions because of the particularly interest rate sensitive nature of the banking business (Flannery and James, 1984; Staikouras, 2003 and 2006; Hahm, 2004). Amongst this line of thought, fundamentalists' approach is theory of Efficient Market Hypothesis (EMH) which has been put forward by Fama (1971) who further categorizes these markets on the basis of their reaction to the information fed to them in weak, semi-strong, or strong form of markets. But the Popular Model Theory shares a different view point altogether, it is nothing but the qualitative explanation of price which suggests that people act incongruously to the information that they receive and freely available information is not necessarily already incorporated into a stock price as EMH attests, which we can say is quite similar to Keynes view point.



A country's financial sector consists of its money market and capital market. Both of them are not only highly correlated but also interdependent. Since the investors for both are same thus they look for the best opportunity wherever available, thus rates in one market do affect investments in another. An important rate which further acts as a barometer for determining other rates in the market is bank rate, the rate at which RBI lends to other banks. The above figure depicts the trend of sensex and the interest rate in India for a year i.e. January 2015 to December 2015.

Review of Literature

Relationship of interest rate and stock returns has been widely examined by researchers. In literature, French, K. R., Schwert, G. W. and Stambaugh, R. E. (1987), French, K. R., Schwert, G.W. and Stambaugh, R. E. (1987) documented theoretically, that stock returns responded negatively to both the long term and short term interest rates. Choi and Jen (1991) report that the expected returns on common stocks are systematically related to the market risk and the interest-rate risk. The findings of the study indicate that the interest-rate risk for small firms is a significant source of investors' portfolio risk and the interest-rate risk for large firms is "negative". Allen and Jagtianti (1997) pointed out that the interest rate sensitivity to stock returns has decreased dramatically since the late 80's and the early 90's because of the invention of interest rate derivative contracts used for hedging purposes. The empirical results of Muradoglu and Metin (2001) indicate that growth rates of interest rates negatively affect stock returns with a significant lag in short run dynamic model. Few recent studies carried out in Indian context, Bhanumurthy and Agarwal (2003) observed that nominal interest rates adjust only to movements in the wholesale market prices but the relationship was not robust. They concluded that interest rate determination in India need not focus much on the domestic inflation rate, as there seems to be no strong co-movement between them.

Bhatt and Virmani (2005) showed that short term interest rates in India are getting progressively integrated with those in the US even though the degree of integration is far from perfect.

Research Gap

According to modern financial theory, systematic factors affect the long run return on financial asset. It implies that security market must have a significant relationship with real and financial sectors of the economy. This relationship is generally viewed in two ways. (i) The macro economic variables influence the stock market operations especially equity returns.(ii) Similarly, the stock market is influenced by the macroeconomic variables. Although, there is much research about short run and long run relationship between macroeconomic variables and stock markets especially in developed countries, but there is a gap in literature in this area of developing and fast developing economies. The present study is necessitated to fill up the gap. The present study is aimed to evaluate and analyze the Interest rate and Stock returns in India.

Objectives of the Study

The specific objectives of the study are as follows;

- To identify the nature of relationship between interest rate and stock returns in India .
- To examine the causal nexus between interest rate and stock returns in India .

Hypothesis

In order to answer whether the interest rate causes the stock returns, the following hypotheses are framed:

- Ho: Interest rate does not cause stock returns
- Ho: Stock returns does not cause interest rate

Data

The data used for the present research work is secondary data. The data were collected from BSE and Reserve Bank of India (RBI). The relevant literature was gleaned from books, journals and magazines. The impact of Interest rate and Stock returns in India is studied, by

using daily data from January 2015 to December 2015. This period is not only the economic growth period but also depicted a great deal of volatility.

Variable Justification

Dependent Variable

Stock Returns

Stock returns has been calculated by using following equation

$$R_t = \ln(I_t / I_{t-1})$$

Where,

R_t = Return for month 't'

I_t and I_{t-1} = daily BSE-Sensex Index for months 't' and t-1 respectively.

BSE SENSEX is taken as a proxy for equity returns, also-called the **BSE 30** or simply the **SENSEX**, is a free-float market-weighted stock market index of 30 well-established and financially sound companies listed on Bombay Stock Exchange. The 30 component companies which are some of the largest and most actively traded stocks, are representative of various industrial sectors of the Indian economy. Published since 1 January 1986, the S&P BSE SENSEX is regarded as the pulse of the domestic stock markets in India. The base value of the S&P BSE SENSEX is taken as 100 on 1 April 1979, and its base year as 1978–79. On 25 July 2001 BSE launched DOLLEX-30, a dollar-linked version of S&P BSE SENSEX. As of 21 April 2011, the market capitalisation of S&P BSE SENSEX was about ₹29733 billion (US\$442 billion) (47.68% of market capitalisation of BSE), while its free-float market capitalisation was ₹15690 billion (US\$233 billion). During 2008-12, Sensex 30 Index share of BSE market capitalisation fell from 49% to 25% due to the rise of sectoral indices like BSE PSU, Bankex, BSE-Teck, etc. The SENSEX is the benchmark index of the Indian Capital Markets with wide recognition among individual investors, institutional investors, foreign investors and fund managers. The BSE SENSEX is not only systematically designed but also based on globally accepted construction and review methodology. First compiled in 1986, SENSEX is a basket of 30 constituent stocks representing a sample of large, liquid and representative companies.

Independent Variables

Interest rate

When reference is made to *the Indian interest rate* this often refers to the repo rate, also called the key short term lending rate. If banks are short of funds they can borrow rupees from the Reserve Bank of India (RBI) at the repo rate, the interest rate with a 1 day maturity. If the central bank of India wants to put more money into circulation, then the RBI will lower

the repo rate. The reverse repo rate is the interest rate that banks receive if they deposit money with the central bank. This reverse repo rate is always lower than the repo rate. Increases or decreases in the repo and reverse repo rate have an effect on the interest rate on banking products such as loans, mortgages and savings. Lending Rates of Daily call money rates have been used as proxy of interest rate.

Methodology The present study tried to expose the relationship between interest rate and stock returns, to focus on ‘causality’ among the variables using the method developed by Granger. Statistical and econometric tools have been used to test and verify the results of the study for their accuracy. The tools namely Augmented Dickey Fuller Test (for checking the stationarity of the data and finding unit roots in time series) and Granger causality test (for verifying the causal nexus between the Interest rate and Bombay Stock Exchange) have been used for examining the short-run interdependence between variables.

Empirical Results

At first, Augmented Dickey Fuller test is used to find out the stationarity of the data set. The results of the test are given in the Table 01 below. The data on stock returns i.e BSE Sensex and interest rate are stationary at first difference which rejects the null hypothesis that data has a unit root.

Table:1

Unit Root Test Results

Variables	Augmented Dickey Fuller Test
	At First Difference (P-Value)
Stock Returns	0.0000*
Interest Rate	0.0000*

*At 5% level of significance

After determining the stationarity of the data set, Granger Causality test developed by Granger and which are generally adopted for examining the short-run inter-dependence between variables, the same have been applied in the present study too. to answer whether changes in interest rate cause changes in Stock returns or changes in Stock returns cause changes in interest rate, applying order one *i.e.* I (1). Table 02 represents the empirical results of Granger causality test between interest rate and stock returns.

Table :2

Null Hypothesis	Lags	F-statistics	Probability	Results
Interest rate does not cause Stock Returns	2	2.25346	0.0973	ACCEPTED
Stock Returns does not cause Interest rate	2	2.40676	0.0924	ACCEPTED

The test results depicts that, the study fails to reject the null hypothesis of interest rate does not cause stock returns as well as the null hypothesis of Stock returns does not cause interest rate. This implies that the interest rate neither affects Stock returns nor stock returns affect the interest rate. The study indicates that the interest rate does not cause the Stock returns. It is consistent with the results of Mok (1993), Gjerde and Saettem (1999), Mukherjee and Naka (1995) and Humpe and Macmillan, (2009). The results was also consistent with the findings of many researcher is India like Bhattacharya and Mukherjee (2002), Pratik and Vina (2004) and the results is contrary to Mukhopadhyay and Sarkar (2003), Debabrata Mukhopadhyay and Nityananda Sarkar (2003), Ray, Pranthik and Vani Vina (2004),Lakahmi R. Nair (2008), Shahid Ahmed (2008). The Stock returns does not cause interest rate. It is consistent with the results of Bhattacharya and Mukherjee (2002), Shahid Ahmed (2008) and the result is contrary to Ratanapakorn and Sharma (2007). Thus, it is observed that, stock market has no relation with the growth of Interest rate in India and vice versa for the selected period of time.

Conclusions Based on various alternative specifications, we have arrived on a parsimonious specification which is based on macroeconomic theories as well as is useful for generating forecasts. The current study attempted to assess the causality association between Interest rate and Sensex. The findings reveal that no causality is noticed between Interest rate

and share returns. In fact, econometric models were applied to get more accuracy to the analyse and thereby the results bring to the fore that there is no such causality association between them. Therefore, the ultimate conclusion is that the Interest rate never influences share returns and vice versa. Thus, the study pragmatically proves that the Interest rate has no inducement on the stock returns and in turn stock returns has no influence over the interest rate for the selected period of time for one year.

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